卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-ENG-ECI-DD-GL-0109 30-B003701S-Z0109

第 1 页 共 1 页
PAGE OF

图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

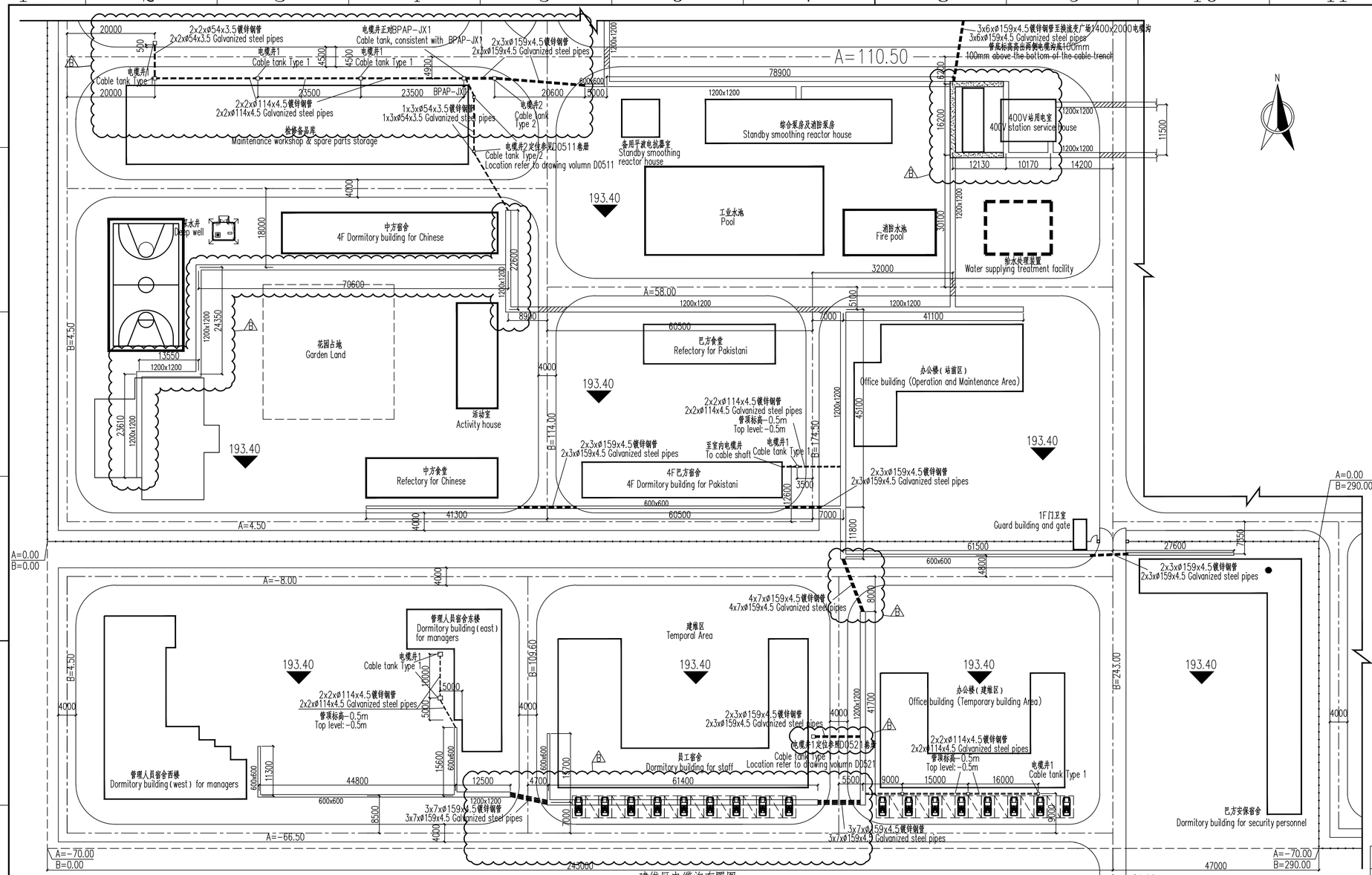
总图 专业 第 01 卷 第 09 册
General Layout SPECIALITY VOLUME BOOK

卷册名称 站前区与建维区电缆沟(第一次修改)
Volume name Cable Trenches for Temporal Area (Version B)

批 准 APPROVED BY	张志强
审 核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	刘恪
图 纸 3 张 DRAWING 3 NOS.	打印件 0 本 PRINTED COPIES 0 NOS.

序号 No.	图 号 DRAWING NO.	文件名称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
1	30-B003701S-Z0109-01	站前区与建维区电缆沟布置图 General layout of cable trenches for temporal area	B	1		
2	30-B003701S-Z0109-02	电缆沟详图 Construction details of cable trenches	B	1		
3	30-B003701S-Z0109-03	涵洞详图 Construction details of culverts	A	1		

注：在本次卷册修改中，序号01、02图为本次升版图纸。



建维区电缆沟布置图

Layout of cable trenches for temporal Area

图例：
Legend:

- 建筑物 Building
- 围墙 Enclosing wall
- 道 路 Road
- 地下结构 Undergroud structure
- 电缆沟 Cable trench
- 电缆沟盖板封闭（上铺20mm水泥砂浆） Trench cover sealed by 20mm cement mortar
- 沟渠 Cable trench
- 2x3xø159x4.5镀锌钢管 2x3xø159x4.5 Galvanized steel pipes
- 2x2xø114x4.5镀锌钢管 2x2xø114x4.5 Galvanized steel pipes

场地说明:

Notes for construction site:

1.本图采用巴基斯坦UTM-43N平面坐标系，EGM2008高程系统。
1.Coordinate system: Pakistan UTM-43N. Elevation system: EGM2008
2.相对平面坐标以米（m）为单位，尺寸以毫米（mm）为单位。
2.Relative coordinates system is measured in meter. Dimensions are measured in millimeters.
3.场地平均设计标高为193.40m。
3.Site level is equal to 193.40m.
4.本图建筑（A、B）坐标与UTM-43N平面坐标系关系如下所示:
4.The relationship between relative coordinate system and UTM-43N system is as below:

5.本工程初步设计尚未审定，未通过NIDC审核，最终方案还存在修改的可能性，施工图请联系业主与设计确认，由此引起的方案变更设计不予负责。
5.Because of preliminary design documents are not approved by NIDC, the project solution has prossibility to be modified. The design contractor has no responsibility for the project solution changed, if the construction contractor not get authorized by project owner and design contractor before construction started.

电缆沟说明:

Notes for construction of cable trenches:

1.电缆沟设计使用年限为50年。
1.The design service limit is 50 years.
2.电缆沟均以中心线为定位。
2.All the cable trenches are located by center axis.
3.电缆沟盖板设计活荷载 4kN/m。
3.The design live load on trench cover is 4kN/m.^{2}
4.混凝土强度采用fc'=28Mpa，钢筋ASTM A615 Grade 60, fy = 420MPa。垫层采用100厚fc'=21Mpa混凝土
4.The strength of concrete used in foundation is fc'=28Mpa,reinforcement ASTM A615 Grade 60, fy = 420MPa. The strength of concrete used in cushion is fc'=21Mpa.
5.电缆沟及盖板的混凝土保护层厚度如下表:
5.The thickness of concrete protection is as following table:

部位	保护层	
侧壁	内侧	19.1mm
	外侧	50.8mm
底板	上侧	19.1mm
	下侧	76.2mm
盖板	下侧	19.1mm

Items	Concrete protection	
Side wall	Interior	19.1mm
	Exterior	50.8mm
Base slab	Top	19.1mm
	Bottom	76.2mm
Cover	Bottom	19.1mm

6.电缆沟底纵向排水坡度 i=0.5%，每隔30米左右设一集水坑，集水坑用埋管通向附近雨井。
6.The longitute slope for drainage at bottom is i=0.5%. A sump pit should be set at each 30m long, and connected to the manhole nearby by pipe.
7.过道路电缆沟及预留管二侧各伸出路宽500mm，所有埋管顶标高低于-0.5m。
7.The culverts and embedded pipes should be extended for 500mm long over the each side of the road. The top level of all embedded pipes should be lower than -0.5m.
8.过道路电缆沟与电缆沟连接处设25mm沥青缝，填充材料采用沥青麻丝，表面材料采用中性硅酮耐候密封胶。
8.A 25mm wide joint should be set at the connection of cable trench and culvert. The joint should be filled with asphalt flax inside and finished by neutral silicone weather sealant.
9.电缆沟每隔15米设一道25mm伸缩缝，填充材料采用新型聚氨酯类材料填充，表面材料采用中性硅酮耐候密封胶。
9.A 25mm wide joint should be set at each 15m long of cable trench. The joint should be filled with polyurethane materials inside and finished by neutral silicone weather sealant.
10.外露铁件热镀锌防腐；现场焊接破坏时采取防腐措施，底漆：环氧富锌，中漆：云铁氯化橡胶，面漆：氯化橡胶丙烯酸磁漆。
10.All the exposed steel members should be hot-galvanized. The steel member welded in site should be paint coating for anti-corrosion. Priming coating: Epoxy zinc rich paint; Second coating: Ferric chloride chlorinated rubber paint; Finishing coating: Chlorinated rubber acrylic enamel paint.
11.电缆沟需采用好土回填，并分层夯实，压实系数不小于0.94。
11.The soil under cable trenches should be good quality soil and tamped by layers, the compacting factor should be more than 0.94.
12.电缆沟压顶梁上铺设30mmx10mm橡胶垫。
12.The 30mmx10mm rubber pad should be equipped on the cappings of trench.
13.地下混凝土要求:
1）水胶比不大于0.45；
2）最大氯离子含量（水泥用量的百分比）：0.10；测试方法参《ASTM C 1218M》。
3）水泥型号：采用抗硫酸盐的硅酸盐水泥，水泥型号需满足《ACI 318-14》表19.3.2.1对应于暴露等级S1的水泥种类要求（ASTM C150 或ASTM 595 或ASTM C1157对应的类别）；
4）水泥中掺入矿物掺合料（矿物掺合料的掺量、使用方法和耐腐蚀性能按相应产品的使用说明并经验证后确定）。
13.Requirements for Underground Concrete:
1) Maximum water-cement ratio is 0.45;
2) Maximum water-soluble chloride ion content is 0.1;Test method refers to "ASTM C 1218M".
3) Type of Concrete: Concrete should be made of sulfate-resistance silica cement.Cement type should fulfill the requirements for concrete used in exposure grade S1, according to《ACI 318-14》table 19.3.2.1.(ASTM C150,ASTM 595 or ASTM C1157 corresponding cement type).
4) Add mineral admixtures (the dosage, using method and anti-corrosion ability should be tested according to the product manual)
14.600x600电缆沟过道路预埋2x3xø159x4.5镀锌钢管，管底标高高出电缆沟内底50mm。
14.2x3xø159x4.5 galvanized steel pipes should be embedded at where 600x600 type trench crossing the road. The bottom level of pipes should be 50mm above the trench bottom.
15.与建筑物相连的电缆沟定位供参考，以建筑结构图纸为准。
15.The location of cables which connectes to building is reference only, and it will subject to the structural drawings of building.
16.电缆井定位可根据现场实际情况调整，需保证电缆井的间距小于20m。
16.The location of the cable tank can be adjustted on site according to the situation, the distance of each cable tank should be less than 20m.
17.电缆埋管管底标高高于电缆沟底50mm，施工方请电缆沟壁预留开孔。
17.The bottom level of pipes which connects to cable trenches should be 50mm above the trench bottom. The construction team should reserve openning on side wall of cable trenches for pipes.

变电一次	郑煜	2019.12.11
专 业	会 签	日 期
Speciality	Sign	Date

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes
B	刘艳	2019.12.11	Adjust cable trenches, cable tanks and embedded pipes.

巴基斯坦默蒂亚里-拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - G L - 0 1 0 9 - 0 0 1 B

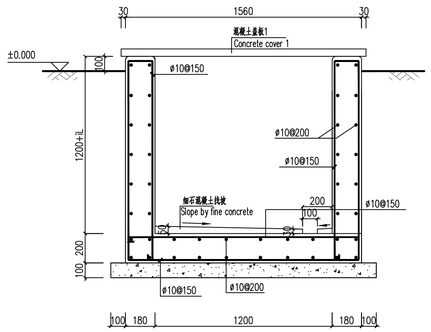
	姓名 Name	日期 Date
设计 Designed	刘艳	2019.12.11
校核 Checked	周峰	2019.12.11
审核 Reviewed	吕建清	2019.12.11
批准 Approved	杨涛	2019.12.11

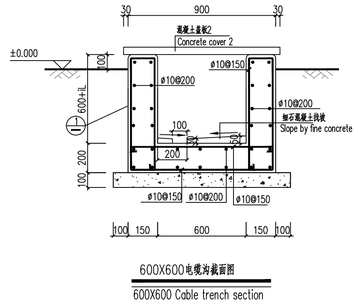
站前区与建维区电缆沟布置图

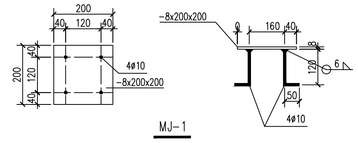
General layout of cable trenches for temporal area

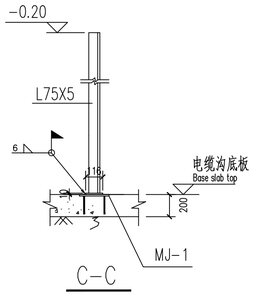
比例 Scale: 1:500 | 图号 DWG NO.: 30-B003701S-Z0109-001 | 版本 REV.: B

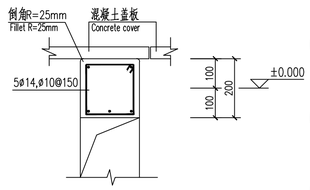
594X841

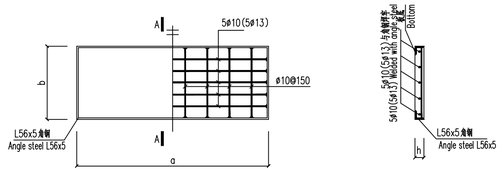








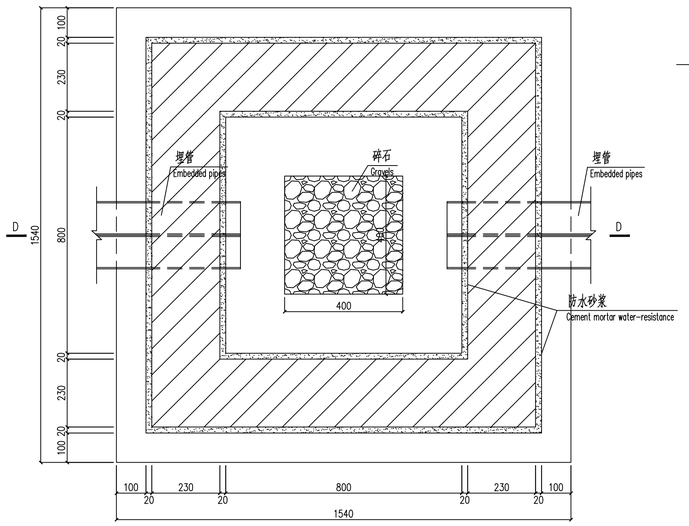


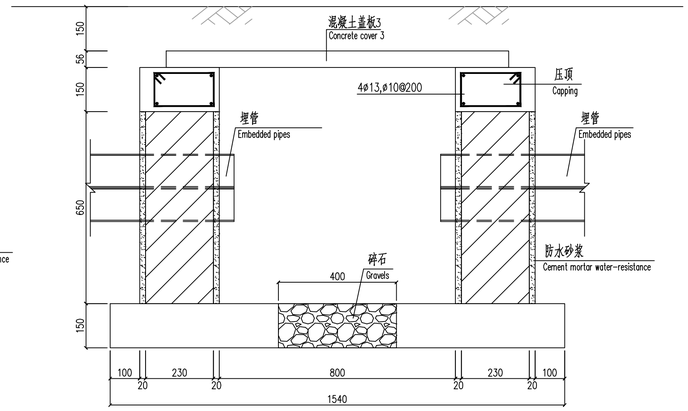


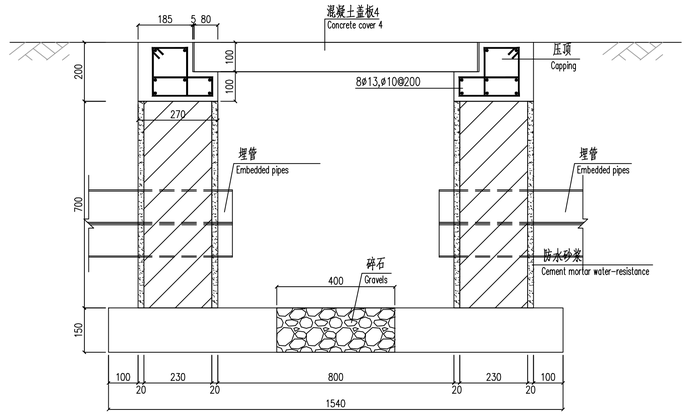
混凝土盖板1
Concrete cover

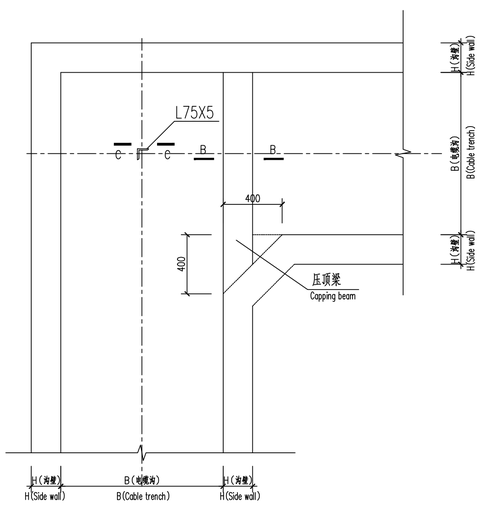
电缆沟盖板一览表
Concrete cover list

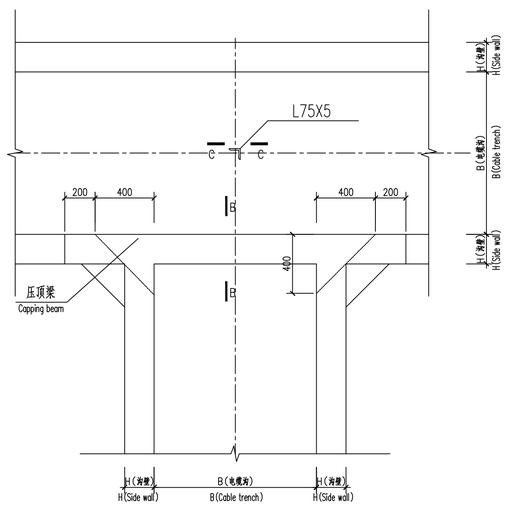
序号 No.	沟净宽 Width	盖板长度a Parameter a	盖板宽度b Parameter b	盖板厚度 Thickness h	主筋 Rebars	备注 Remarks
1	1200	1620	496	56	5ø10	
2	600	960	496	56	5ø10	
3	800	1160	496	56	5ø10	Cable Tank Type 1
4	800	960	496	100	5ø13	Cable Tank Type 2











Reference Drawings

Drwg. No.	Title

Revisions			
B	刘惟	2019.12.11	Add a new type of cable tank
	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
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拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - G L - 0 1 0 9 - 0 0 2 B

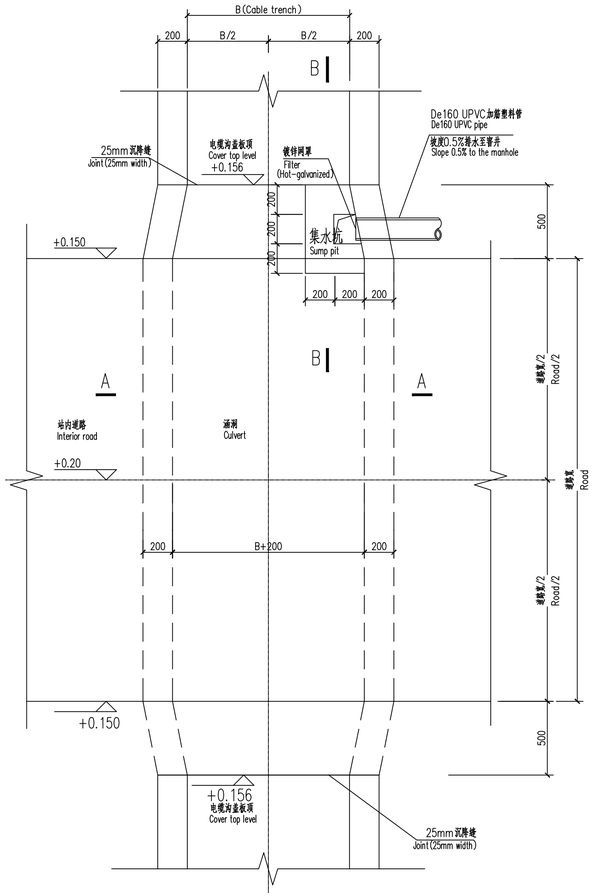
	姓名 Name	日期 Date
设计 Designed	刘惟	2019.12.11
校核 Checked	周必强	2019.12.11
审核 Reviewed	毛德强	2019.12.11
批准 Approved	杨少彬	2019.12.11

电缆沟详图
Construction details of cable trenches

比例 Scale | 图号 DWG NO. 30-B003701S-Z0109-002 | 版本 REV. B

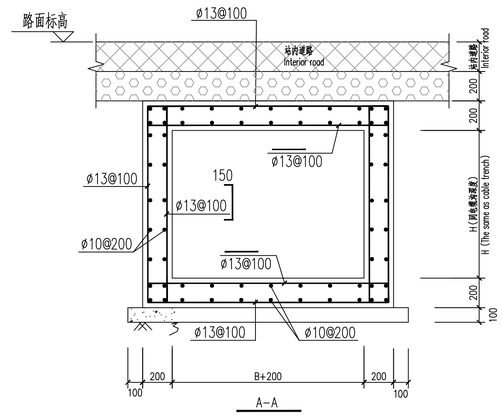
专业 Speciality	会签 Sign	日期 Date
变电一次	郑煜	2019.12.11

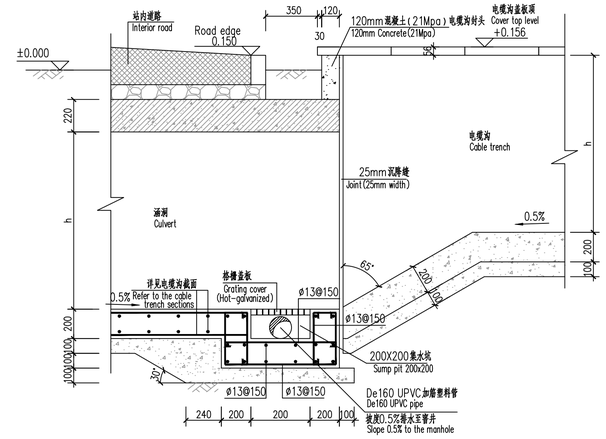
594X841



涵洞平面图

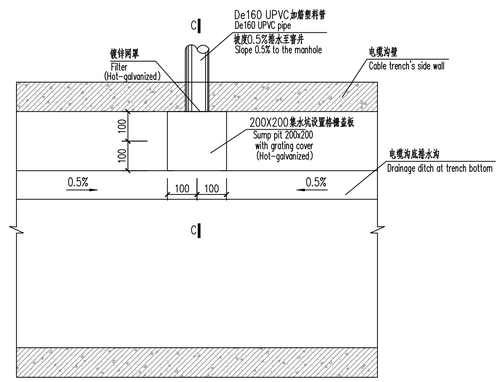
Plan view of culvert





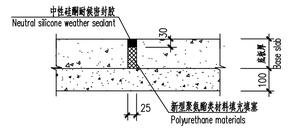
B-B 涵洞与电缆沟连接详图

B-B Conection details between culvert and cable trench



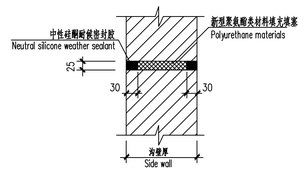
电缆沟集水井详图

Details of sump pit



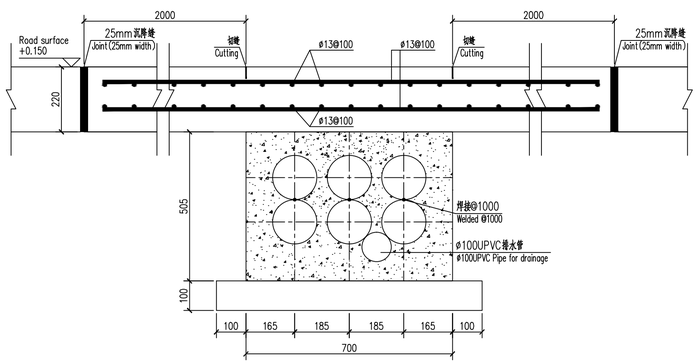
沟底伸缩缝示意图

Details of connection joint at base slab



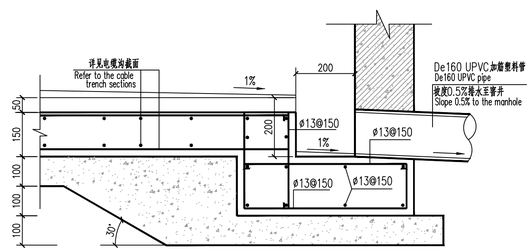
沟壁伸缩缝详图

Details of connection joint at side wall



过道路处理管详图

Construction details of joint (Embedded pipes crossing road)



C - C

变电一次	郑煜	2018.02.12
专 业	会 签	日 期
Speciality	Sign	Date

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里-拉合尔±660kV直流输电工程

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拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - G L - 0 1 0 9 - 0 0 3 A

	姓名 Name	日期 Date
设计 Designed	刘恺	2018.02.12
校核 Checked	闫峰	2018.02.12
审核 Reviewed	邹德清	2018.02.12
批准 Approved	韩松涛	2018.02.12

过道路电缆沟涵洞详图

Construction details of culverts

比例 Scale | 图号 DWG NO. 30-B003701S-Z0109-003 | 版本 REV. A

594X841